Convex Order and Moment Persuasion*

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Abstract

We study Bayesian persuasion with multidimensional states when the sender's payoff depends on the distribution of posterior means. The choice of information policy in such an environment reduces to an optimization over distributions of posterior means dominated by the prior in convex order. We study extreme points of this feasible set using results on irreducible-component structure from martingale transport theory. When the prior assigns zero mass to the union of the relative boundaries of the non-singleton components, then on each such component the distribution of posterior means is dominated by the prior in convex order. In such cases extreme points must satisfy affine independence of the conditional posterior-mean support in almost every component. Without boundary negligibility, local affine independence remains necessary but is not sufficient. Unlike in the one-dimensional case, in environments with multidimensional states, extreme points can have uncountably many lower-dimensional components that are not singletons.

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1 Introduction

In moment persuasion, a sender chooses an optimal signal to maximize expected payoff when the objective depends on posterior beliefs through their means. The set of feasible signals has a useful convex-order representation: the distributions of posterior means induced by signals are the mean-preserving contractions of the prior. This connection follows from the martingale characterization of convex order in Strassen (1965) and the comparison-of-experiments perspective of Blackwell (1953). After Rothschild and Stiglitz (1971, 1978) introduced this tool to economics, it has recently been revived to link the persuasion framework of Kamenica and Gentzkow (2011) to the analysis of posterior means in Kolotilin et al. (2017), Dworczak and Martini (2019), and Arieli et al. (2023).

In a moment persuasion model with a compact convex state space and a bounded upper semicontinuous indirect payoff, the sender essentially maximizes an affine upper semicontinuous functional over the set of the mean-preserving contractions of the prior which is a compact convex set. Bauer's maximum principle then implies that at least one optimal posterior-mean distribution is an extreme point. This observation motivates a geometric question: which mean-preserving contractions are extreme? An answer restricts the distributions that need to be considered in optimization, although extremality by itself does not establish optimality for a particular payoff.

In one dimension, integrated-distribution inequalities identify intervals on which information can be pooled and points at which it is fully revealed. This structure underlies the characterizations in Kleiner et al. (2021) and Arieli et al. (2023). In several dimensions, pooling can occur on sets of different dimensions, and a partition can contain uncountably many nontrivial components. The irreducible-component structure developed by De March and Touzi (2019), together with the related analysis of Obłój and Siorpaes (2017), provides a way to organize these possibilities. For a given pair of measures in convex order, every martingale coupling sends a posterior mean only to states in the convex closure of its component. We use this existing decomposition as the starting point for the extreme-point

analysis.

Theorem 1 provides a complete geometric characterization of mean-preserving contractions in multidimensional settings. We show that whenever a measure μ is dominated by μ_0 in convex order, the state space admits a canonical partition into convex, relatively open cells such that the restriction of μ to each cell remains dominated by the corresponding restriction of μ_0. Within each cell, any feasible martingale transport must map states into the closure of that same cell, a concentration property that parallels the irreducible-component structure identified in the martingale transport literature. This decomposition generalizes the one-dimensional partition into intervals and singleton points and provides the geometric backbone for our analysis of extreme points.

To obtain a sharp and tractable characterization as Theorem 1, we impose a mild regularity condition—Assumption 1—which requires that the relative boundaries of the convex cells be negligible under the prior. Assumption 1 does not directly impose any assumption on the prior. Though absolute continuity of the prior does not guarantee this assumption in general (because the convex partition may contain uncountably many cells), this assumption still holds in many economically relevant environments, including those where the partition arises from the subdifferential of a convex function, or in two-dimensional settings where Lebesgue measurability ensures boundary negligibility.

Theorem 2 characterizes extremality under the same boundary condition. A feasible distribution is extreme if and only if its conditional support is affinely independent in almost every component. Thus a component of dimension d contains at most $d + 1$ conditional posterior means, while a singleton component reveals its state. The proof first establishes the result on a single component by constructing a bounded signed perturbation and controlling it uniformly by the convex-order slack. A measurable disintegration argument then passes from the local statements to the global characterization.

Section 4 examines the limits of this characterization when Assumption 1 fails. In such cases, the relative boundaries of the convex cells may carry positive prior mass,

allowing martingale transports to exploit boundary interactions and mix mass across otherwise distinct concentration regions. Although a version of the local affine-independence condition remains necessary for extremality, it is no longer sufficient: boundary points introduce additional degrees of freedom that can break global extremality even when each local component is well-behaved. This analysis clarifies the precise role of Assumption 1 in our geometric characterization and highlights the subtleties that arise in multidimensional persuasion when boundary effects cannot be ruled out.

The distinction between local and global support matters for economic applications. A distribution of posterior means may have finite support within each component and still have an uncountable aggregate support. Our product-recommendation example has a uniform prior on the unit square and payoff $U(x) = x_1 x_2$. Revealing the sum of the two coordinates induces the unique optimal posterior-mean distribution. This distribution is nonatomic, yet each of its uncountably many one-dimensional pooling components contains a single posterior mean. An explicit payoff identity and a convex-majorant certificate verify optimality and uniqueness.

The paper extends the results of Kleiner et al. (2024), who show that every finitely supported extreme mean-preserving contraction of an absolutely continuous prior admits a convex partition such that, on each cell, the mean-preserving contraction has affinely independent support. In addition to this necessary condition, they provide a sufficient criterion for testing whether a given finitely supported mean-preserving contraction is extreme. Our contribution is to establish a necessary and sufficient characterization of extremality under Assumption 1. This assumption is substantially weaker than Kleiner et al. (2024)'s assumptions of finite support and absolute continuity of the prior, since we allow more general prior and infinite support. Moreover, Example 3 demonstrates that our convex partition can be strictly coarser than theirs, implying that extremality may require affine independence on larger sets than those identified by their partition.

This paper also complements the dual approach of Dworczak and Kolotilin (2024). They develop a persuasion duality using optimal prices and their contact sets to characterize

optimal signals and the associated pooling regions. Their Theorem 8 establishes that, under differentiability of the sender's objective and continuity of the prior, there exists an optimal signal whose posterior means within each price-induced pooling region are extreme points of their own closed convex hull. In multiple dimensions, however, this condition permits infinitely many posterior means within a single region. They explicitly suggest that a finer partition derived from martingale transport might yield a bound of (d+1) posterior means per region, leaving this question for future research. Our characterization provides a conditional resolution of this proposed refinement: whenever an extreme optimal distribution satisfies our boundary assumption, its canonical components satisfy the stronger requirement of affine independence. More broadly, our criterion applies to feasible distributions independently of the sender's objective or the construction of an optimal dual price.

Our paper contributes to the growing literature on the extreme-point approach in economic theory (e.g., Yang and Zentefis (2024), Lahr and Niemeyer (2024), Yang and Yang (2025)). Among these studies, the most closely related is Yang and Yang (2026), which establishes connections between stochastic optimization, extreme-point decompositions, and order-preserving couplings. Their equivalence theorem identifies closure of the test-function cone under pointwise minima as the key condition for a general pointwise decomposition theorem, with applications to multidimensional mean-preserving spreads and stochastic dominance. However, the class of convex functions that characterises mean-preserving contractions is not closed under pointwise minima. Consequently, their general decomposition framework does not apply to the feasible set studied in this paper.

Section 2 develops the conditional decomposition and the boundary assumption. Sections 3 and 4 study extremality with and without that assumption. Section 5 derives the implementation result from the polar-set theorem of De March and Touzi (2019): in a single component with negligible boundary, polar sets are exactly those covered by two marginal null strips. A countable-mixture argument then yields a coupling whose conditional supports equal the support of the prior almost everywhere. Section 6 applies the results to moment persuasion.

2 Characterization

2.1 Preliminaries and Notation

For a set $A \subset \mathbb{R}^n (n \geq 1)$, let $ri(A)$ denote its relative interior, $rbd(A)$ denote its relative boundary, $cl(A)$ denote its closure, and $conv(A)$ denote its convex hull. For a probability measure μ on $\mathbb{R}^n$, let $supp(\mu)$ denote its support, and $\hat{supp}(\mu) := cl(conv(supp(\mu)))$ be the smallest convex closed set containing support of μ. δ_x denotes a Dirac measure concentrated on $x \in \mathbb{R}^n$. For measurable $A \subset \mathbb{R}^n$, μ_A denotes the restriction of μ to A.

Unless otherwise stated, $X = \Omega$ is a nonempty compact convex subset of $\mathbb{R}^n$. We say a collection of subsets of X, $\{A_x\}_{x \in X}$, is a **convex partition** of X if (1)For any $x \in X$, $x \in A_x$ and A_x is a convex set, (2) $\cup_{x \in X} A_x$=X, (3) For any $x, y \in X$, either $A_x = A_y$, or $A_x \cap A_y = \emptyset$.

We mainly consider the convex partition $\{A_x\}_{x \in X}$ of X with each A_x being relatively open in this paper. The measurability of $\{A_x\}_{x \in X}$ means measurability of the label map $K(x) = A_x$ into the space of nonempty relatively open convex subsets, endowed with its Wijsman topology.[1] Analytically measurable maps are sufficient for this paper, but a Borel version can be chosen. If $\lambda = K\#\mu$, meaning $\lambda(A) = \mu(\{x : K(x) \in A\})$ for a measurable $A \subset \{A_x\}_{x \in X}$, then disintegration gives

$$\mu = \int_{\{A_x\}_{x \in X}} \mu_{|A_x} \lambda(dA_x)$$

$\mu_{|A_x}(\{y : K(y) = A_x\}) = 1$ λ-almost everywhere. This defines conditional probabilities even when every individual cell has zero μ-mass. If $\mu(A_x) > 0$, the conditional probability for that cell is $\frac{\mu_{A_x}}{\mu(A_x)}$.

Consider two probability measures μ and μ_0 on X with finite first-order moment, we say μ and μ_0 are in the **convex order**, written as $\mu \preccurlyeq \mu_0$, if $\int \varphi d\mu \leq \int \varphi d\mu_0$ for all convex

[1]The set $\hat{\mathcal{K}}$ of all convex closed subsets of X is a Polish space when endowed with the Wijsman topology, and it can be shown that $\hat{\mathcal{K}}$ is isomorphic to $ri\hat{\mathcal{K}} := \{riK : K \in \hat{\mathcal{K}}\}$, see De March and Touzi (2019) for an explanation of the Wijsman topology

functions $\varphi : X \to \mathbb{R}$. Let $MPC(\mu_0)$ denote the set of all probability measures that are dominated by μ_0 in the convex order. The following standard compactness property follows from the fusion framework of Elton and Hill (1992).

Lemma 1 *If $\Omega \subset \mathbb{R}^n$ is compact and convex and $\mu_0 \in \Delta(\Omega)$, then $MPC(\mu_0)$ is a nonempty convex compact subset of $\Delta(\Omega)$ in the topology of weak convergence.*

Define $\Pi(\mu, \mu_0) := \{p \in \Delta(X \times \Omega) \mid p_X = \mu, p_\Omega = \mu_0\}$, which contains all the possible probability measures on $X \times \Omega$ given the marginal on X is μ and the marginal on Ω is μ_0. The subset of **martingale transports** between μ and μ_0 is

$$\mathcal{M}(\mu, \mu_0) := \{p \in \Pi(\mu, \mu_0) \mid \mathbb{E}[Y \mid Z] = Z \text{ for } (Z, Y) \sim p\}.$$

Notice that by Strassen (1965), $\mathcal{M}(\mu, \mu_0)$ is nonempty if and only if $\mu \in MPC(\mu_0)$.

For any given $x \in X$ and $p \in \mathcal{M}(\mu, \mu_0)$, the conditional kernel p_x is defined μ-a.e. by

$$p(dx, d\omega) = \mu(dx) p_x(d\omega)$$

A finite collection $\{x_1, ..., x_k\} \subset \mathbb{R}^n$ is **affinely independent** if the unique solution to $\sum_{i=1}^k \beta_i x_i = 0$ and $\sum_{i=1}^k \beta_i = 0$ is $\beta_1 = \beta_2 = ... = \beta_k = 0$. A set of affinely independent vectors contains at most $n + 1$ elements.

2.2 Geometric Characterization

To motivate the multidimensional characterization, recall the elegant one-dimensional description of mean-preserving contractions. Assume $\Omega = [0, 1]$, a distribution $G \in \Delta([0, 1])$ is a mean-preserving contraction of $F \in \Delta([0, 1])$ if and only if $\int_0^x G(x)dx \leq \int_0^x F(x)dx$ for all $x \in [0, 1]$ and equality holds at $x = 1$.

If F is a continuous distribution, accoridng to the proof of Proposition 1 in Arieli et al. (2023), these inequalities and equalities partition the state space into "local" intervals and

singleton points:

- On each "local" interval $(a, b) \subset [0, 1]$, $\int_a^b G(x)dx = \int_a^b F(x)dx$, while $\int_a^c G(x)dx < \int_a^c F(x)dx$ for all $c \in (a, b)$;
- At each singleton x, $\int_0^x G(x)dx = \int_0^x F(x)dx$.

Based on this "local" structure, Arieli et al. (2023) prove that the extremality of G implies the support of G contains at most 2 points on each "local" interval.

In higher dimensions, no direct analogue of these integral constraints exists. The challenge is therefore to identify an appropriate multidimensional analogue of these "local" intervals. This motivates the paper of De March and Touzi (2019) on the martingale transport, which provides the structural foundation for our analysis.

Proposition 1 *(De March and Touzi 2019) There exists $\hat{p} \in \mathcal{M}(\mu, \mu_0)$ such that for all $p \in \mathcal{M}(\mu, \mu_0)$, $su\hat{p}p(p_x) \subset su\hat{p}p(\hat{p}_x)$ for μ-a.s. x. Furthermore $su\hat{p}p(\hat{p}_x)$ is μ -a.s. unique, and we may choose this kernel so that:*

- *$K : x \to su\hat{p}p(\hat{p}_x)$ is analytically measurable $X \to \hat{\mathcal{K}}$,*
- *$x \in A_x := risu\hat{p}p(\hat{p}_x)$ for all $x \in X$, and $\{risu\hat{p}p(\hat{p}_x), x \in X\}$ is a convex partition of Ω.*

Proposition 1 states that the state space can be partitioned into convex cells such that every martingale transport must map each state x into the closure of its own cell. These are the multidimensional analogues of the "local intervals" in the one-dimensional case, which called irreducible components by De March and Touzi (2019). A direct corollary of Proposition 1 is that μ coincides with μ_0 on all A_x that are singletons and not on the relative boundary of other cells of the convex partition.

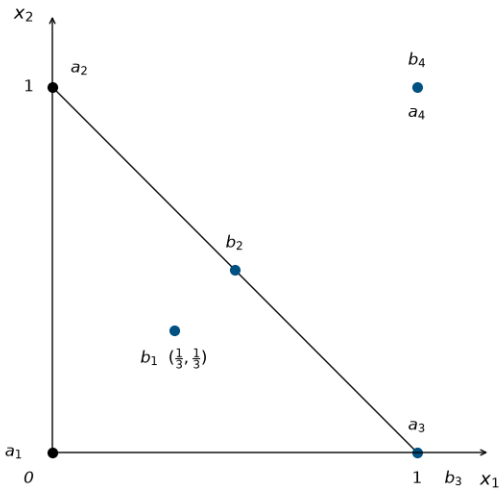
Corollary 1 $\mu_{|\{|X\setminus cl(A)\}} = \mu_{0|\{X\setminus cl(A)\}}$, *where* $A := X\setminus\{x \in X : A_x = \{x\}\}$.

Proof: By Proposition 1, $\{A_x\}$ form a convex partition of X such that $\hat{p}_x(cl(A_x)) = 1$ for μ-a.s. x. If $x \in X\backslash A$, then $A_x = \{x\}$ so that $\hat{p}_x = \delta_x$. If $x \in A$, then $A_x \subset A$ so that $supp(\hat{p}_x) \subset cl(A)$.

If $B \subset X$ is a Borel set, then $\mu_0(B) = \int_X \mu(dx)\hat{p}_x(B) = \int_A \mu(dx)\hat{p}_x(B \cap cl(A)) + \int_{X\backslash A} \mu(dx)\delta_x(B)$. Moreover, if $B \subset X\backslash cl(A)$, $\mu_0(B) = \int_{X\backslash A} \mu(dx)\delta_x(B) = \mu(B)$ ■

If $n = 1$ and μ_0 is a continuous probability measure, then in μ_0-a.s. sense, $X\backslash cl(A)$ is set of singleton points, and all the non-singleton A_x are exactly the "local" intervals. The following example illustrates what the multidimensional analogues of the "local intervals" can look like in two dimensions.

Figure 1: Example 1



Example 1 *Let* $X = [0, 1]^2$*, and let*

$$a_1 = (0, 0),\ a_2 = (0, 1),\ a_3 = (1, 0),\ a_4 = (1, 1),$$
$$b_1 = (1/3, 1/3), \quad b_2 = (1/2, 1/2), \quad b_3 = a_3, \quad b_4 = a_4.$$

μ_0 *assigns masses* $1/12, 5/24, 11/24, 1/4$ *at* a_1, a_2, a_3, a_4*, respectively, and* μ *assigns*

mass $1/4$ *at each* b_i*. The unique martingale coupling has kernels*

$$p_{b_1} = \tfrac{1}{3}(\delta_{a_1} + \delta_{a_2} + \delta_{a_3}), \quad p_{b_2} = \tfrac{1}{2}(\delta_{a_2} + \delta_{a_3}), \quad p_{b_3} = \delta_{a_3}, \quad p_{b_4} = \delta_{a_4}.$$

The non-singleton cells are

$$A_{b_1} = \mathrm{ri}\,\mathrm{conv}\{a_1, a_2, a_3\}, \qquad A_{b_2} = \mathrm{ri}\,\mathrm{conv}\{a_2, a_3\}.$$

The remaining points may be assigned singleton cells. The singleton a_4 *lies outside the closure of the non-singleton region, so* $\mu(\{a_4\}) = \mu_0(\{a_4\})$*. In contrast,* a_3 *is a boundary point of both displayed cells and* $\mu(\{a_3\}) \neq \mu_0(\{a_3\})$*.*

To simplify our language in describing the local structure of all $\mu \in MPC(\mu_0)$, we introduce the following definition.

Definition 1 *Given a convex and relatively open set* $A \subset X \subset \mathbb{R}^n$*,* $\mu_{|A}$ *is* ***concentrated*** *on* $cl(A)$ *if*

- *for all* $p \in \mathcal{M}(\mu, \mu_0)$ *and for* $\mu_{|A}$*-a.s.* x*,* $\hat{supp}(p_x) \subset cl(A)$*,*
- *there exists* $\hat{p} \in \mathcal{M}(\mu, \mu_0)$ *such that* $\hat{supp}(\hat{p}_x) = cl(A)$ *for* $\mu_{|A}$*-a.s.* x*.*

In particular, if there is only one n-dimensional concentration region, we say μ is concentrated on X if $\mu_{|ri(X)}$ is concentrated on X.

To streamline the geometric characterization, we introduce a mild regularity condition ensuring that the relative boundaries of the convex cells in the partition are negligible. Recall that Proposition 1 provides a convex partition $\{A_x\}_{x \in X}$ of the state space, where each $A_x = ri(\hat{supp}(\hat{p}_x))$ for a canonical martingale transport $\hat{p}$. The potential difficulty arises when the relative boundary of one cell intersects with relative boundary of another cell, creating additional degrees of freedom for martingale transports and complicating the analysis of extreme points. As an illustration consider point a_3 in example 1, there exists a

martingale transport(in fact, it is the unique element in $\mathcal{M}(\mu, \mu_0)$ provided above) mapping it to b_1, b_2, b_3, each of which belongs to different concentration regions.

To rule out these boundary-interaction pathologies, we define $\mathcal{V} := \cup_{\{x \in X: A_x \neq \{x\}\}} rbd(A_x)$, where $\{A_x\}_{x \in X}$ is defined in Proposition 1, and we introduce the following assumption:

Assumption 1 *$\mathcal{V}$ is μ_0-null set.*

Assumption 1 means the union of all relative boundaries of non-singleton cells has zero μ_0-measure. This assumption eliminates the possibility that a point lies simultaneously on the boundary of multiple convex cells in a way that affects feasible martingale transports. We delay our more detailed discussion of Assumption 1 to section 2.3.

Under Assumption 1 we can combine Proposition 1 and Corollary 1 to obtain a simple geometric characterization of convex order as follows.

Theorem 1 *Suppose Assumption 1 holds, then $\mu \in MPC(\mu_0)$ if and only if there exists a collection of disjoint convex and relatively open sets $\{A_x\}_{x \in X}$ such that for μ-a.s. x,*

- $x \in A_x$,
- $\mu_{|A_x}$ *is concentrated on* $cl(A_x)$,
- $\mu_{|A_x} \preccurlyeq \mu_{0|A_x}$,
- $\mu_{|\{\Omega \setminus A\}} = \mu_{0|\{\Omega \setminus A\}}$, *where* $A = \cup_{\{x \in X: A_x \neq \{x\}\}} A_x$.

Proof: Necessary condition: According to Proposition 1, $\mu_{|A_x}$ is concentrated on $cl(A_x)$ and $rbd(A_x)$ is μ_0-null set, it is trivial $\mu_{|A_x} \preccurlyeq \mu_{0|A_x}$. Define A_i as each non-singleton A_x that does not lie on the relative boundary of another cell in Proposition 1. If a non-singleton A_x lies on the relative boundary of another cell, then it is μ_0- and $\mu-$ null set given Assumption 1 holds. If $A_x = \{x\}$ and lies on the relative boundary of another cell, then it is either zero measure under both μ_0 and μ or lies on the relative interior of some cell A_i. So the only

A_x matters here is the singleton that does not lie on the relative interior of other cells. According to Corollary 1, $\mu_{|\{\Omega\setminus A\}} = \mu_{0|\{\Omega\setminus A\}}$.

Sufficient condition: For any convex function $\phi : X \to R$,

$$\begin{aligned}\int_X \phi d\mu &= \int_{X\setminus\cup_{I\in I}A_i} \phi d\mu + \int_{\cup_{I\in I}A_i} \phi d\mu \\ &\leq \int_{X\setminus\cup_{I\in I}A_i} \phi d\mu_0 + \int_{\cup_{I\in I}A_i} \phi d\mu_0 \\ &= \int_X \phi d\mu_0\end{aligned}$$

■

Theorem 1 shows that convex order admits a local decomposition: the global dominance is equivalent to convex dominance on each convex cell of the canonical partition, together with equality outside the non-singleton cells. This mirrors the one-dimensional structure, where the state space decomposes into intervals and singleton points. This local structure is the key to the characterization of extreme points in Section 3.

2.3 The Generality of Assumption 1

Before turning to the characterization of extreme points in Section 3, we discuss the generality of Assumption 1, which plays a central role in our geometric decomposition. Since each A_x in the convex partition is a convex set, and since the relative boundary of any convex set has Lebesgue measure zero (Lang (1986)) one might expect Assumption 1 to hold whenever μ_0 is absolutely continuous with respect to Lebesgue measure. However, the convex partition may contain uncountably many convex sets, and Lebesgue measure is not countably additive over uncountable unions. Consequently, the set $\mathcal{V}$ need not be Lebesgue measurable, and even when it is, it may fail to have measure zero. Our Lemma 2 gives an affirmative answer when $n = 1, 2$ and a negative result for $n \geq 3$.

Lemma 2 *For $n = 1$ and any given convex partition $\{A_x\}_{x\in X}$, $\mathcal{V} := \cup_{\{x\in X: A_x\neq\{x\}\}} rbd(A_x)$ is zero Lebesgue measure. For $n = 2$ and any given convex partition $\{A_x\}_{x\in X}$, if $\mathcal{V}$ is*

Lebesgue measurable, then $\mathcal{V}$ is zero Lebesgue measure. For $n \geq 3$, there exists a convex partition $\{A_x\}_{x \in X}$ such that $\mathcal{V}$ is compact and has positive Lebesgue measure.

In one dimensional case, convex partition consists of at most countable intervals with each of their boundaries being the two endpoints. Thus $\mathcal{V}$ is zero Lebesgue measure. In two dimensional case, the only problematic cells are one-dimensional ones (line segments). This is because there can be at most countably many two-dimensional cells, and their boundaries contribute zero Lebesgue measure.[2] For the one-dimensional cells, Larman (1971)'s first lemma on disjoint line segments ensures that the union of their endpoints has measure zero. The proof of Lemma 2 carefully handles cases where closures of linear segments intersect, showing that one can enlarge the set of endpoints to a disjoint family of line segments whose endpoints still form a measure-zero set. In contrast, for $n \geq 3$, Larman (1971) shows that one can construct a compact family of disjoint line segments whose set of endpoints has positive Lebesgue measure. Treating all other points as singleton cells yields a convex partition for which $\mathcal{V}$ is compact and has positive measure. This establishes that Assumption 1 may fail in higher dimensions.[3]

It is well-known that the projection of a piecewise linear convex function onto its domain creates a convex partition of state space. By Rademacher's Theorem (Theorem 10.8 in Villani (2008)), we know the subset of the nondifferentiable domain of a convex function is Lebesgue-negligible. This motivates Lemma 3, which provides another way to verify the Assumption 1. Given a convex function $f(x)$ defined on X, denote $ND_f := \{x \mid f \text{ is nondifferentiable at } x\}$,

Lemma 3 *If there exists a convex function f such that $\mathcal{V}$=$ND_f \cup \partial X$, then $\mathcal{V}$ is zero Lebesgue measure.*

Kleiner et al. (2024) consider a special convex partition named power diagram which is

[2]Since $\mathbb{R}^n$ is separable, any family of pairwise disjoint nonempty open subsets is at most countable (see Thm. 2.8 in Oxtoby (2013)). Because each full-dimensional convex cell contains a nonempty open set (its interior), the convex partition has at most countably many n-dimensional cells.

[3]It is still an open question whether this construction can be realized as a convex partition that satisfies the concentration property for some μ and μ_0 that are in the convex order.

essentially a projection of a piecewise linear convex function with each cell of the projection having same dimensions as X. This is a special case of Lemma 3 since the boundary of each cell in the power diagram is the nondifferentiable region of the piecewise linear convex function plus the boundary of X.

3 Extreme Points

We say a probability measure μ is finitely supported if $supp(\mu)$ is finite. Let $FMPC(\mu_0)$ denote the set of all finitely supported probability measures that belong to $MPC(\mu_0)$. We begin by characterizing the extreme points of $FMPC(\mu_0)$ in the case where all mass lies within a single concentration region.

Proposition 2 *Suppose $\mu \in FMPC(\mu_0)$ is concentrated on X, and Assumption 1 holds, then μ is an extreme point of $FMPC(\mu_0)$ only if μ has affinely independent support.*

Proof: Suppose the support of μ is not affinely independent. Let $\{x_i \mid i \in I\}$ denote the support of μ. According to Corollary 2 in Section 5, $\exists p \in \mathcal{M}(\mu, \mu_0)$, $\forall x_i$, $supp(p_{x_i}) = supp(\mu_0)$. Let $\mu_i = \mu(x_i)p_{x_i}$, then $\mu_i(X) = \mu(x_i), r(\mu_i) =: \frac{1}{\mu_i(X)} \int_X x d\mu_i(x) = \int_X x dp_{x_i}(x) = x_i$, and $\mu_0 = \sum_i \mu_i$. The remainder of the proof follows the argument of Proposition 1 in Kleiner et al. (2024).[4] ■

The Proposition 1 in Kleiner et al. (2024) consider a convex partition which is the finest one preserving convex dominance on each cell. Theorem 1 implies that convex dominance also holds on each cell of our (coarser) convex partition.[5] Proposition 2 therefore shows that within each cell, affine independence is still necessary for extremality of finitely supported measures.

[4]Notice the procedure of constructing graphs in the proof of Proposition 1 in Kleiner et al. (2024) always stops with a graph G_N where $\{x_i \mid i \in C\}$ is affinely dependent, because $supp(p_{x_i}) = supp(\mu_0)$ for all $i \in I$, $\{x_i \mid i \in I\}$ are all included in C in the end

[5]Depending on the given μ_0 and μ, our convex partition can be strictly coarser than the convex partition in Kleiner et al. (2024). Example 3 in Appendix provides an illustration.

We can extend Proposition 2 to any extreme points of $MPC(\mu_0)$ given there exists only one concentration region. The proof is in the Appendix.

Proposition 3 *Suppose $\mu \in MPC(\mu_0)$ is concentrated on X, and Assumption 1 holds, then μ is an extreme point of $MPC(\mu_0)$ only if μ has affinely independent support.*

Proposition 3 is proved by a two-sided perturbation argument. If $\operatorname{supp}\mu$ is affinely dependent, we construct a nonzero signed measure $\sigma = h\mu$, with bounded h, that preserves total mass and first moments. For infinite support, this construction uses positive-mass neighborhoods rather than individual atoms. The crucial step uses the concentration and boundary assumptions to establish a uniform estimate

$$\int_X \varphi \, d(\mu_0 - \mu) \geq c \left| \int_X \varphi \, d\sigma \right|,$$

where $c > 0$ is independent of the continuous convex function φ. Thus, one sufficiently small $\varepsilon > 0$ makes both $\mu \pm \varepsilon\sigma$ feasible, contradicting extremality. Although this shares the mass-redistribution intuition of the proof of Proposition 1 in Kleiner et al. (2024), their proof relies on finitely many positive-mass atoms, terminating overlap-graph constructions, and transfers along finite spanning trees. These steps do not directly apply to nonatomic distributions, whose individual support points carry zero mass. The additional argument in Proposition 3 is therefore to construct a perturbation of the measure itself and control all convex-order inequalities simultaneously.

Finally, combining proposition 3 with Theorem 1, we can give our full characterization of extreme points of $MPC(\mu_0)$. The proof of Theorem 2 further shows the nuance of localization structure of any mean-preserving contractions. For the sufficiency proof, we essentially prove that for any two mean-preserving contractions, their convex combination must have a coarser convex partition than theirs. For the necessity proof, If a given probability measure $\mu \in MPC(\mu_0)$ is an extreme point of $MPC(\mu_0)$, then its conditional probability measure to any cell A_x in the convex partition is an extreme point of $MPC(\mu_{0|A_x})$. It allows us to focus on each simpler local mean-preserving contraction.

Theorem 2 *Suppose $\mu \in MPC(\mu_0)$ and Assumption 1 holds, then μ is an extreme point of $MPC(\mu_0)$ if and only if there exists a collection of disjoint convex and relatively open sets $\{A_x\}_{x\in X}$ such that for μ-a.s. x,*

- $x \in A_x$,
- $\mu_{|A_x}$ *is concentrated on* $cl(A_x)$,
- $\mu_{|A_x}$ *has affinely independent support,*
- $\mu_{|A_x} \preccurlyeq \mu_{0|A_x}$,
- $\mu_{|\{\Omega\setminus A\}} = \mu_{0|\{\Omega\setminus A\}}$, *where* $A = \cup_{\{x\in X: A_x \neq \{x\}\}} A_x$.

Proof: Sufficient condition: If μ is not a extreme point, then we can find two probability measures $\mu_1, \mu_2 \in MPC(\mu_0)$, such that $\mu = \frac{1}{2}\mu_1 + \frac{1}{2}\mu_2$. Define corresponding convex partitions $\{A_x^1\}_{x\in X}$ and $\{A_x^2\}_{x\in X}$ for μ_1 and μ_2 as in Proposition 1. We say μ_1 has finer concentration regions than μ with respect to μ_0 if $A_x^1 \subset A_x$ for μ_1-a.s x. According to Lemma 7 in Appendix, both μ_1 and μ_2 have finer concentration regions than μ with respect to μ_0. The supports of μ_1 and μ_2 must be subsets of support of μ in each A_x. μ has affinely independent supports in each A_x. Because convex dominance implies that means must coincide for all these measures on given A_x, then $\mu_{|A_x} = \mu_{1|A_x} = \mu_{2|A_x}$, which implies $\mu = \mu_1 = \mu_2$.

Necessary condition: We only need to prove the third condition, all other conditions come from Theorem 1. According to Proposition 1, the partition we have comes from a measurable mapping, so we have the following disintegration form,

$$\mu = \int_{\{A_x\}_{x\in X}} \mu_{|A_x} \lambda(dA_x),$$

where each $\mu_{|A_x}$ is a conditional probability measure defined on the corresponding convex component, $\lambda(A') = \mu(\{x : A_x \in A'\})$ for all measurable $A' \subset \{A_x\}_{x\in X}$. The extremality of μ requires the extremality of $\mu_{|A_x}$ for λ-almost all A_x. If A_x is singleton, the third condition

is trivial. Then applying Proposition 3 to all A_x that are not singletons, we finish the proof.

■

4 When Assumption 1 does not hold

Assumption 1 ensures that the relative boundaries of the non-singleton cells in the convex partition are negligible under the prior. This regularity is what allows the global convex-order relation to decompose cleanly into independent local dominance relations on each cell. When Assumption 1 fails, this clean separation may break down: boundary points may simultaneously belong to the closures of multiple cells, and martingale transports may exploit these boundary interactions to mix mass across otherwise distinct concentration regions. As a result, the geometric characterization in Theorem 1 no longer holds in full generality, and the structure of extreme points becomes more subtle.

Even without Assumption 1, Proposition 1 still provides a canonical convex partition $\{A_x\}_{x\in X}$ of the state space, and Corollary 1 continues to identify the fully revealing singleton cells. What fails is the guarantee that the closures of distinct non-singleton cells are disjoint up to μ_0-null sets. Consequently, a mean-preserving contraction may assign mass to boundary points that lie in the closures of multiple cells, creating additional feasible martingale transports that do not respect the local concentration structure.

Lemma 4 and Theorem 3 establish what can still be said about extreme points of mean-preserving contraction set when Assumption 1 is not satisfied.

Lemma 4 *Suppose $\mu \in MPC(\mu_0)$ and $\mu_{|ri(X)}$ is concentrated on X, then μ is an extreme point of $MPC(\mu_0)$ only if μ has affinely independent support on $ri(X)$.*

Proof: If $\mu(ri(X)) = 0$, the conclusion is immediate. Hence assume $\mu(ri(X)) > 0$. By concentration, there exists a martingale coupling $\pi(dx, dy) = \mu(dx)p_x(dy)$ between μ and

μ_0 such that $su\hat{p}p(p_x) = X$ for μ-almost every $x \in ri(X)$. Define probability measures

$$\beta_0 := \int_{ri(X)} p_x \mu_{|ri(X)}(dx).$$

Then $\mu_{|ri(X)}(dx)p_x(dy)$ is a martingale coupling between $\mu_{|ri(X)}$ and β_0, so $\mu_{|ri(X)} \preceq \beta_0$.

We first show that extremality of μ in MPC(μ_0) implies extremality of $\mu_{|ri(X)}$ in $MPC(\beta_0)$. For any $\beta \preceq \beta_0$, define

$$\mu_\beta := \mu(ri(X))\beta + \mu_{rbd(X)}.$$

This is a probability measure. For every convex function $\varphi : \mathbb{R}^d \to \mathbb{R}$, Jensen's inequality gives

$$\begin{aligned}\int_X \varphi \, d\mu_\beta &\leq \mu(ri(X)) \int_X \varphi \, d\beta + \int_{X\setminus ri(X)} \varphi(x) \, \mu(dx) \\ &\leq \int_{ri(X)} \int_X \varphi(y) \, p_x(dy) \, \mu(dx) + \int_{X\setminus ri(X)} \int_X \varphi(y) \, p_x(dy) \, \mu(dx) \\ &= \int_X \varphi \, d\mu_0.\end{aligned}$$

Thus $\mu_\beta \in$ MPC(μ_0). If $\mu_{|ri(X)} = \frac{1}{2}\beta_1 + \frac{1}{2}\beta_2$, $\beta_1, \beta_2 \in$ MPC(β_0), then $\mu = \frac{1}{2}\mu_{\beta_1} + \frac{1}{2}\mu_{\beta_2}$. Extremality of μ gives $\mu_{\beta_1} = \mu_{\beta_2}$, and $\mu(ri(X)) > 0$ then implies $\beta_1 = \beta_2$. Hence $\mu_{|ri(X)}$ is extreme in MPC(β_0).

To apply Proposition 3, we construct an intermediate prior η with

$$\mu_{|ri(X)} \preceq \eta \preceq \beta_0, \qquad \eta(rbd(X)) = 0,$$

whose canonical irreducible component is $ri(X)$, ν-almost everywhere.

Let Z have distribution $\mu_{ri(X)}$, and Y have distribution $\mu_{|ri(X)}(dx)P_x(dy)$. and let T be independent of (Z, Y) and uniformly distributed on $(0, 1)$. Define $W := (1-T)Z+TY$ and η as the distribution of W. Since $\mathbb{E}[Y \mid Z] = Z$, independence gives $\mathbb{E}[W \mid Z] = \frac{1}{2}Z + \frac{1}{2}\mathbb{E}[Y \mid$

$Z] = Z$. Therefore $\mu_{|ri(X)} \preceq \eta$. For every convex function $\varphi : \mathbb{R}^d \to \mathbb{R}$,

$$\begin{aligned}\int_X \varphi \, d\eta &= \mathbb{E}[\varphi((1-T)Z + TY)] \\ &\leq \tfrac{1}{2}\mathbb{E}[\varphi(Z)] + \tfrac{1}{2}\mathbb{E}[\varphi(Y)] \\ &\leq \mathbb{E}[\varphi(Y)] = \int_X \varphi \, d\beta_0,\end{aligned}$$

where the last inequality uses $\mu_{|ri(X)} \preceq \beta_0$. Hence $\eta \preceq \beta_0$.

Because $Z \in ri(X)$ almost surely and $(1-t)x + ty \in ri(X)$ for $x \in ri(X)$, $y \in X$, $0 < t < 1$, we have $\eta(ri(X)) = 1$, and consequently $\eta(\operatorname{rbd} X) = 0$.

Let Q_x be the conditional distribution of W given $Z = x$. Then $Q(dx, dw) := \mu_{ri(X)}(dx)Q_x(dw)$ is a martingale coupling between $\mu_{ri(X)}$ and η. We claim that $\operatorname{supp} p_x \subseteq \operatorname{supp} Q_x$ for $\mu_{ri(X)}$-almost every x. Fix such an x, a point $y \in \operatorname{supp} P_x$, and $\varepsilon > 0$. Choose $\delta \in (0, 1)$ such that $\delta \operatorname{diam}(X) < \varepsilon/2$. Whenever $y' \in X \cap B(y, \varepsilon/2)$ and $t \in (1-\delta, 1)$,

$$\|(1-t)x + ty' - y\| \leq (1-t)\|x - y'\| + \|y' - y\| < \varepsilon.$$

It follows that

$$Q_x(X \cap B(y, \varepsilon)) \geq \delta \, P_x(X \cap B(y, \varepsilon/2)) > 0.$$

Thus $y \in \operatorname{supp} Q_x$, proving the claim. Consequently, $\hat{supp}(Q_x) = X$ for ν-almost every x. By maximality of the canonical conditional convex supports, the canonical irreducible paving associated with (ν, η) therefore satisfies $A_x = ri(X)$ for ν-almost every x.

All the assumptions of Proposition 3 now hold for (ν, η). Moreover, $\nu \in \operatorname{MPC}(\eta) \subseteq \operatorname{MPC}(\beta_0)$. Since ν is extreme in $\operatorname{MPC}(\beta_0)$, it is also extreme in $\operatorname{MPC}(\eta)$. Proposition 3 implies that $\operatorname{supp}(\nu)$ is affinely independent. Finally, since $ri(X)$ is relatively open in $\operatorname{aff} X$, $\operatorname{supp}(\mu) \cap ri(X) = \operatorname{supp}(\nu) \cap ri(X)$. Therefore μ has affinely independent support on $ri(X)$.

■

If μ has more than one concentration region, then we can apply Lemma 4 to each

concentration region.

Theorem 3 *$\mu \in MPC(\mu_0)$ is an extreme point of $MPC(\mu_0)$ only if there exists a collection of disjoint convex and relatively open sets $\{A_x\}_{x\in X}$ such that for μ a.s. x,*

- *$x \in A_x$,*
- *$\mu_{|A_x}$ is concentrated on $cl(A_x)$,*
- *$\mu_{|A_x}$ has affinely independent support,*
- *$\mu_{\{\Omega\setminus cl(A)\}} = \mu_{0\{\Omega\setminus cl(A)\}}$, where $A := \mathbb{R}^n\setminus\{x \in \mathbb{R}^n : A_x = \{x\}\}$.*

Proof: We only need to prove $\mu_{|A_x}$ has affinely independent support for all $x \in X$, the other conditions come directly from Proposition 1 and Corollary 1. According to Proposition 1, the partition we have come from a measurable mapping, so we have the following disintegration form,

$$\mu = \int_{\{A_x\}_{x\in X}} \mu_{|A_x}\lambda(dA_x),$$

where each $\mu_{|A_x}$ is a conditional probability measure defined on the corresponding convex component, $\lambda(A') = \mu(\{x : A_x \in A'\})$ for all measurable $A' \subset \{A_x\}_{x\in X}$. The extremality of μ requires the extremality of $\mu_{|A_x}$ for λ-almost all A_x. Then applying Lemma 4 to all A_x, we finish the proof. ■

Theorem 3 therefore provides necessary conditions for extremality even when Assumption 1 fails. However, these conditions are no longer sufficient. The failure of boundary negligibility allows martingale transports to mix mass across the closures of distinct cells, and this additional flexibility can destroy extremality even when each local component satisfies affine independence. The following example illustrates this phenomenon.

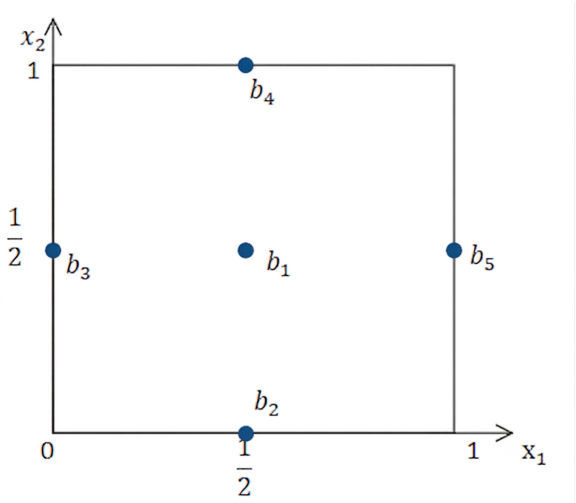
Example 2 *Let $X = [0, 1]^2$, U is a uniform probability measure on X, and let*

$$b_1 = (1/2, 1/2), \quad b_2 = (1/2, 0), \quad b_3 = (0, 1/2),$$
$$b_4 = (1/2, 1), \quad b_5 = (1, 1/2).$$

Define

$$\mu_0 = \frac{1}{5}U + \frac{1}{5}\sum_{i=2}^{5}\delta_{b_i}, \qquad \mu = \frac{3}{5}\delta_{b_1} + \frac{1}{10}\sum_{i=2}^{5}\delta_{b_i}.$$

Figure 2: Example 2



μ *and* μ_0 *are in the convex order. The support of* μ *includes 5 points,* b_1,b_2, b_3, b_4, b_5. *Each of them belongs to a different cell of the convex partition:*

$$A_{b_1} = (0,1)\times(0,1), A_{b_2} = \{b_2\}, A_{b_3} = \{b_3\}, A_{b_4} = \{b_4\}, A_{b_5} = \{b_5\}.$$

Assumption 1 fails in the example, because relative boundary of A_{b_1} *includes points which have positive measure. Although support of* $\mu_{|A_{b_i}}$ *is affinely independent for all i=1,2,3,4,5,* $\mu = \frac{1}{2}\delta_{(\frac{1}{2},\frac{1}{2})} + \frac{1}{2}\mu_1$, *where* $\delta_{(\frac{1}{2},\frac{1}{2})}$ *is a Dirac measure concentrated on* $(\frac{1}{2},\frac{1}{2})$ *and* μ_1 *assigns 0.2 to all* $b_i, i = 1,2,3,4,5$. *It is easy to verify both* $\delta_{(\frac{1}{2},\frac{1}{2})}$ *and* μ_1 *are mean-preserving contractions of* μ_0. *Thus* μ *is not an extreme point.*

This example shows that without Assumption 1, affine independence within each cell is no longer sufficient for extremality. Boundary interactions introduce additional degrees of freedom that cannot be ruled out by local geometric conditions alone.

5 Implementations

In this section, we examine the implementation of mean-preserving contractions via martingale transports. We say $p \in \Delta(X \times \Omega)$ implements some $\mu \in MPC(\mu_0)$ if $p \in \mathcal{M}(\mu, \mu_0)$. A Borel set $N \subseteq X \times X$ is $\mathcal{M}(\mu, \mu_0)$-*polar* if $p(N) = 0$ for every $p \in \mathcal{M}(\mu, \mu_0)$. Polar sets describe pairs of posterior means and states that no feasible implementation can map between. The following theorem reveals the remarkable flexibility of martingale transports:

Theorem 4 *Suppose that $\mu \in MPC(\mu_0)$ is concentrated on X and $\mu_0(\text{rbd}\, X) = 0$. A Borel set $N \subseteq X \times X$ is $\mathcal{M}(\mu, \mu_0)$-polar if and only if there are Borel sets $N_\mu, N_0 \subseteq X$ with $\mu(N_\mu) = \mu_0(N_0) = 0$ such that*

$$N \subseteq (N_\mu \times X) \cup (X \times N_0)\,^{6}.$$

In other words, the only polar sets are those that lie nontrivially outside the cartesian product of supports of μ and μ_0. Any set intersecting both supports of μ and μ_0 in a nontrivial way can be transported by some feasible martingale transport. One particularly important consequence of Theorem 4 is that it ensures that there exists a martingale transport whose conditional distribution spans the entire support of μ_0 for μ-almost every x. The existence of this particular martingale transport is the backbone for the proof of Proposition 3.

Corollary 2 *Under the hypotheses of Theorem 4, there is $p \in \mathcal{M}(\mu, \mu_0)$ such that*

$$\text{supp}(p_x) = \text{supp}(\mu_0) \qquad \textit{for } \mu\textit{-almost every } x.$$

If μ is finitely supported, the equality holds at every point of $\text{supp}(\mu)$.

Proof. Let $S = \text{supp}(\mu_0)$ and choose a countable base $(U_k)_{k \geq 1}$ of nonempty relatively open subsets of S. Each U_k has positive μ_0-measure. Fix k. For a Borel disintegration of

[6] Theorem 4 comes directly from Theorem 2.5 in De March and Touzi (2019) where their $\{Y \notin J(X)\}$ becomes a μ_0-null set under Assumption 1.

$Q \in \mathcal{M}(\mu, \mu_0)$, set

$$T_k(Q) = \{x : Q_x(U_k) > 0\}, \qquad s_k = \sup_{Q \in \mathcal{M}(\mu,\mu_0)} \mu(T_k(Q)).$$

Choose Q^m with $\mu(T_k(Q^m)) \to s_k$. The countable mixture $\bar{Q} = \sum_{m \geq 1} 2^{-m} Q^m$ is a martingale transport, and $T_k(\bar{Q}) = \bigcup_m T_k(Q^m)$ up to a μ-null set. It follows that $\mu(T_k(\bar{Q})) = s_k$. If $s_k < 1$, the set $B = X \setminus T_k(\bar{Q})$ has positive μ-measure. By Theorem 4, some $R \in \mathcal{M}(\mu, \mu_0)$ satisfies $R(B \times U_k) > 0$. The mixture $(\bar{Q} + R)/2$ then makes T_k strictly larger in μ-measure, a contradiction. Thus $s_k = 1$; denote the resulting transport by P^k.

Now put $p = \sum_{k \geq 1} 2^{-k} P^k$. Its conditional distributions charge every U_k for μ-almost every x, simultaneously for all k. Also $p_x(S) = 1$ almost everywhere, by the second marginal constraint. Hence $\mathrm{supp}(p_x) = S$ almost everywhere. Every support point of a finitely supported μ is an atom, so none can belong to a μ-null exceptional set. ■

6 Moment Persuasion

In moment persuasion, the sender's expected payoff after a signal depends on the posterior belief only through its mean. Let $\Omega \subset \mathbb{R}^n$ be compact and convex, let $\mu_0 \in \Delta(\Omega)$ be the prior, and let $U : \Omega \to \mathbb{R}$ denote the sender's indirect payoff. This formulation includes environments in which the receiver's action depends only on the posterior mean and the sender's payoff depends only on that action. It also allows state-dependent realized payoffs when their conditional expectation has the form $U(\mathbb{E}[\omega \mid S])$.[7]

Following the standard argument in Arieli et al. (2023), it is without loss of generality to restrict attention to the unbiased signals where the signal realization space is the same as the posterior mean space and state space, $S = X = \Omega$. For any unbiased signal, its posterior mean $x := \mathbb{E}[\omega \mid s]$ satisfies $\mathbb{E}[\omega \mid x] = x$. Conversely, any martingale coupling

[7] A finite vector of moments can be treated as the mean of a transformed state. The prior is then the distribution of that transformed state, with the corresponding compactness and integrability assumptions; see Section 4 of Dworczak and Kolotilin (2024).

between a posterior-mean distribution and μ_0 induces an unbiased signal that implements that distribution. Thus the feasible posterior-mean distributions are exactly $MPC(\mu_0)$, and the sender solves

$$\max_{\mu \in MPC(\mu_0)} \int_\Omega U(x)\, \mu(dx).$$

Suppose that U is bounded and upper semicontinuous. By the Portmanteau theorem, $\mu \mapsto \int U\, d\mu$ is upper semicontinuous; it is also affine. Bauer's maximum principle therefore implies that an optimal posterior-mean distribution exists and that at least one optimum is an extreme point of $MPC(\mu_0)$.

This existence statement explains the relevance of the preceding geometric analysis. When the hypotheses of Theorem 2 hold for an extreme maximizer, its conditional distribution of posterior means has affinely independent support within almost every component. Without Assumption 1, Theorem 3 supplies necessary conditions for extremality. Neither extremality nor those necessary conditions alone establish optimality for a specified payoff U. Moreover, the components depend on the candidate distribution and the prior; they do not determine which regions the sender should reveal before the optimization problem is solved.

A singleton component at x forces its conditional state distribution to be δ_x, so the state is revealed whenever that posterior mean occurs. Within a component of dimension $d \geq 1$, affine independence restricts the conditional distribution of posterior means to at most $d + 1$ points. Components of dimension $1 \leq d < n$ correspond to pooling on lower-dimensional sets, a possibility that has no one-dimensional counterpart. These statements concern conditional distributions on the components, which may individually have zero prior probability.

An optimality certificate. A convex majorant gives a direct way to verify a candidate optimum. Let $q : \mathbb{R}^n \to \mathbb{R}$ be a finite convex function with $q \geq U$ on Ω. For every feasible

μ,

$$\int U \, d\mu \leq \int q \, d\mu \leq \int q \, d\mu_0.$$

Hence a feasible μ^* is optimal whenever

$$q = U \quad \mu^*\text{-almost surely}, \qquad \int q \, d\mu^* = \int q \, d\mu_0.$$

This sufficient condition follows directly from convex order. It is the weak-duality argument underlying the moment-persuasion approach of Dworczak and Kolotilin (2024); no general claim of dual attainment is needed here.

Example: product recommendations and infinitely many pooling regions. The following example has a unique optimal distribution of posterior means and uncountably many nontrivial pooling regions. Consider a uniformly-distributed prior specialization of the product-information environment in Section 8.1 of Malamud and Schrimpf (2021).

A platform privately observes $\omega = (v, p) \sim \mu_0 := U([0,1]^2)$, where v is a product's value to a consumer and p is the platform's commission if the consumer purchases. The consumer privately observes a purchase cost $c \sim U([0,1])$, independent of ω. If a purchase occurs, the consumer receives $v - c$ and the platform receives p; otherwise both receive zero. Before the state is realized, the platform commits to a possibly randomized signal based on (v, p). It does not observe c, and its randomization is independent of c.

After signal s, write

$$x_1(s) := \mathbb{E}[v \mid s], \qquad x_2(s) := \mathbb{E}[p \mid s].$$

The consumer purchases if $c \leq x_1(s)$, with the tie having probability zero. Independence implies

$$\mathbb{E}\big[p \, \mathbf{1}_{\{c \leq x_1(s)\}} \mid s\big] = x_1(s) x_2(s).$$

If we consider unbiased signal, then the sender's indirect payoff is $U(x) = x_1 x_2$.[8] The platform solves

$$\max_{\mu \in MPC(\mu_0)} \int_{[0,1]^2} x_1 x_2 \, \mu(dx). \tag{1}$$

Consider the signal $S^* = T := v + p$. Exchangeability of v and p gives the posterior mean

$$m(t) := \big(\mathbb{E}[v \mid T = t], \mathbb{E}[p \mid T = t]\big) = \left(\frac{t}{2}, \frac{t}{2}\right).$$

To prove optimality and identify all optimal posterior means, let S be any feasible signal and write $M_1 := \mathbb{E}[v \mid S]$, $M_2 := \mathbb{E}[p \mid S]$. Since $M_1 + M_2 = \mathbb{E}[T \mid S]$, the identity $4M_1M_2 = (M_1 + M_2)^2 - (M_1 - M_2)^2$ and the conditional variance formula yield

$$\begin{aligned}\frac{1}{4}\mathbb{E}[T^2] - \mathbb{E}[M_1 M_2] &= \frac{1}{4}\left(\mathbb{E}[T^2] - \mathbb{E}[(M_1 + M_2)^2]\right) + \frac{1}{4}\mathbb{E}[(M_1 - M_2)^2] \\ &= \frac{1}{4}\left(\mathbb{E}[T^2] - \mathbb{E}\Big[\big(\mathbb{E}[T \mid s]\big)^2\Big]\right) + \frac{1}{4}\mathbb{E}[(M_1 - M_2)^2].\end{aligned}$$

Since

$$\mathrm{Var}(T \mid s) = \mathbb{E}[T^2 \mid s] - \big(\mathbb{E}[T \mid s]\big)^2,$$

the law of iterated expectations gives

$$\mathbb{E}[\mathrm{Var}(T \mid s)] = \mathbb{E}[T^2] - \mathbb{E}\Big[\big(\mathbb{E}[T \mid s]\big)^2\Big].$$

Consequently,

$$\frac{1}{4}\mathbb{E}[T^2] - \mathbb{E}[M_1 M_2] = \frac{1}{4}\mathbb{E}[\mathrm{Var}(T \mid s)] + \frac{1}{4}\mathbb{E}[(M_1 - M_2)^2]. \tag{2}$$

Here

$$\frac{1}{4}\mathbb{E}[T^2] = \frac{1}{4}\left(\mathbb{E}[v^2] + \mathbb{E}[p^2] + 2\mathbb{E}[vp]\right) = \frac{1}{4}\left(\frac{1}{3} + \frac{1}{3} + \frac{1}{2}\right) = \frac{7}{24}$$

is the payoff under S^*, and $\mathbb{E}[M_1 M_2]$ is the payoff under any signal S. Under $S^* = T$, both

[8] The same optimization problem arises with a receiver who chooses $a \in [0,1]^2$ to maximize $-\mathbb{E}[\|a - \omega\|^2 \mid s]$ and a sender whose payoff is $a_1 a_2$. The receiver chooses $a = \mathbb{E}[\omega \mid s]$.

nonnegative terms in (2) vanish. S^* is therefore optimal, with expected payoff $7/24$.

Equality in (2) holds if and only if $\mathrm{Var}(T \mid S) = 0$ and $M_1 = M_2$ almost surely. Therefore every optimal signal satisfies $(M_1, M_2) = m(T)$ almost surely, and the optimal distribution of posterior means, μ^*, is unique. Because U is continuous, it exposes μ^* in $MPC(\mu_0)$; in particular, μ^* is extreme. Since T has a continuous triangular distribution, $\operatorname{supp}\mu^* = \{(z, z) : 0 \le z \le 1\}$, and μ^* is nonatomic. No signal with finitely or countably many distinct posterior means can attain the optimum. The uniqueness is at the level of posterior means: additional information can leave these means unchanged. For example, revealing both T and $|v - p|$ also induces μ^* by symmetry.

The convex-majorant certificate is particularly simple here:

$$q(x) := \frac{(x_1 + x_2)^2}{4}, \qquad q(x) - U(x) = \frac{(x_1 - x_2)^2}{4}.$$

Thus q is convex, coincides with U on the diagonal, and satisfies $\int q \, d\mu_0 = 7/24$. It is constant, hence affine, along each pool of S^*, so both inequalities in the certificate bind for μ^*.

For $t \in (0, 2)$, the signal $T = t$ pools the states in the segment

$$C_t := \{(v, p) \in [0, 1]^2 : v + p = t\} = \{(u, t - u) : \max\{0, t - 1\} \le u \le \min\{1, t\}\}.$$

A version of the conditional prior given $T = t$ is uniform on C_t, whereas the conditional posterior-mean distribution is $\delta_{m(t)}$. The relatively open segments $A_t := \operatorname{ri} C_t$ are pairwise disjoint. They are also the canonical cells defined in Proposition 1. Let (Z, Y) be any martingale coupling of μ^* and μ_0, so that

$$Z \sim \mu^*, \qquad Y \sim \mu_0, \qquad \mathbb{E}[Y \mid Z] = Z.$$

Recall that $q(x) = \frac{1}{4}(x_1 + x_2)^2$. The martingale property implies $\mathbb{E}[Y_1 + Y_2 \mid Z] = Z_1 + Z_2$.

Since $Z_1 + Z_2$ is measurable with respect to Z, we obtain

$$\begin{aligned}\mathbb{E}[(Y_1 + Y_2)(Z_1 + Z_2)] &= \mathbb{E}[(Z_1 + Z_2)\mathbb{E}[Y_1 + Y_2 \mid Z]] \\ &= \mathbb{E}[(Z_1 + Z_2)^2].\end{aligned}$$

Consequently,

$$\begin{aligned}\mathbb{E}[(Y_1 + Y_2 - Z_1 - Z_2)^2] &= \mathbb{E}[(Y_1 + Y_2)^2] - \mathbb{E}[(Z_1 + Z_2)^2] \\ &= 4\left(\int q \, d\mu_0 - \int q \, d\mu^*\right).\end{aligned}$$

Equality of the two integrals therefore gives $\mathbb{E}[(Y_1 + Y_2 - Z_1 - Z_2)^2] = 0$. The integrand is nonnegative, so $Y_1 + Y_2 = Z_1 + Z_2$ almost surely. Let p_z be the distribution of Y conditional on $Z = z$. Since the coordinates of $m(t)$ sum to t, the preceding almost-sure equality implies $p_{m(t)}(C_t) = 1$ for μ^*-almost every t. Because C_t is closed, $\operatorname{supp} p_{m(t)} \subseteq C_t$ for μ^*-almost every t. The sum-revealing implementation attains all of C_t as its conditional support, so it attains the maximal conditional convex support. The endpoints of these segments lie on the boundary of the square, which has zero μ_0-measure. Thus this example exhibits uncountably many lower-dimensional pooling components while satisfying the required boundary negligibility. Each component has one posterior mean, although the aggregate distribution is nonatomic.

Economically, the platform groups high-value, low-commission products with low-value, high-commission products having the same sum. Expected product value determines the probability of purchase, and the commission determines the platform's payoff if a purchase occurs. The platform wants a high probability of purchase when the commission is high. Pooling opposite types helps achieve this: high-value, low-commission products lend credibility to recommendations that also include low-value, high-commission products. The example also demonstrates why a finite bound on the number of posterior means within each component does not imply a finite bound for the information policy as a whole.

7 Conclusion

The irreducible-component structure connects the global convex-order constraint to local restrictions on posterior means. When component boundaries carry no prior mass, the prior and the posterior-mean distribution share a common distribution over cells, and extremality is equivalent to conditional affine independence. Without boundary negligibility, the same local restriction is necessary but need not be sufficient because boundary mass can be allocated across components.

For moment persuasion, this characterization describes the geometry of at least one extreme maximizer under the usual compactness and upper semicontinuity conditions, whenever the relevant boundary hypothesis holds. It does not replace the comparison of payoffs across feasible distributions. The product-recommendation example illustrates both points: a direct optimality certificate identifies a unique, nonatomic posterior-mean distribution, while the geometric decomposition explains its uncountably many one-dimensional pools.

It would be interesting to investigate the following two questions. First, it would be useful to develop conditions on the primitives that guarantee that Assumption 1 is satisfied. Second, characterizing which extreme points are unique solutions by economically meaningful payoffs would bring the geometric analysis closer to constructive solutions of persuasion problems.

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A Omitted Proofs

Proof of Lemma 2. We assume $X = \mathbb{R}^n$ in this proof, if X is a convex compact subset of $\mathbb{R}^n$, the proof is similar. Since $\mathbb{R}^n$ is separable, any family of pairwise disjoint nonempty open subsets is at most countable (see Thm. 2.8 in Oxtoby (2013)). Because each full-dimensional convex cell contains a nonempty open set (its interior), the convex partition has at most countably many n-dimensional cells.

if $n = 1$, we have at most countable many A_x that are not singleton. The relative boundary of each interval A_x is zero Lebesgue measure. Since Lebesgue measure is additive for countable many sets, $\mathcal{V}$ is zero Lebesgue measure.

Since the proof for $n = 2$ relies on several new established results, we treat this case separately as Lemma 5. See the proof of Lemma 5.

If $n \geq 3$, by Theorem 1 from Larman (1971), there exists a disjoint set L of closed line segments in $\mathbb{R}^n$ such that the Lebesgue measure of the set of end points(the relative boundaries of line segments) is positive, where L(and hence the set of end points of all line segments) is compact. Take the relative interiors of the segments as cells and assign each remaining point of Ω to a singleton cell. Together, these cells form a convex partition and $\mathcal{V}$ is compact, has positive measure. ■

Lemma 5 *Let $\{A_i\}_{i\in I}$ be a pairwise disjoint family of non-singleton, relatively open, convex subsets of $\mathbb{R}^2$. Suppose that*

$$\mathcal{V} := \bigcup_{i\in I} \operatorname{rbd} A_i$$

is Lebesgue measurable. Then

$$\mathcal{L}^2(\mathcal{V}) = 0.$$

Proof of Lemma 5:

Step 1: The two-dimensional members.

Let $X_2 := \{x \in X : \dim A_x = 2\}$. For $x \in X_2$, relative openness is ordinary openness in $\mathbb{R}^2$. Hence $\{A_x\}_{x \in X_2}$ is a pairwise disjoint family of nonempty open subsets of $\mathbb{R}^2$. Such a family is countable, since every nonempty open subset of $\mathbb{R}^2$ contains a point of $\mathbb{Q}^2$, and distinct members contain distinct rational points. Moreover, if $A \subset \mathbb{R}^2$ is convex with nonempty interior,then

$$\mathcal{L}^2(\partial A) = 0$$

according to Lang (1986)

Consequently, $P := \bigcup_{x \in X_2} \operatorname{rbd} A_x$ is a countable union of null sets, so

$$\mathcal{L}^2(P) = 0.$$

Step 2: The one-dimensional members.

Let $\mathcal{A}_1 := \{A_x : \dim A_x = 1\}$, and define

$$E := \bigcup_{A \in \mathcal{A}_1} \operatorname{rbd} A.$$

Every $A \in \mathcal{A}_1$ is, in its affine hull, either an open bounded interval, an open ray, or an entire affine line. We have $\mathcal{V} = P \cup E$. Define the set of T-junction endpoints by $T := \{x \in E : x \in H \text{ for some } H \in \mathcal{A}_1\}$.

We first prove that

$$\mathcal{L}^{2,*}(T) = 0.$$

Step 3: T-junction nullity. Fix $x \in T$. Since the members of $\mathcal{A}_1$ are pairwise disjoint, there is a unique $H_x \in \mathcal{A}_1$ such that $x \in H_x$. We call H_x the host of x. Since $x \in E$, choose one $A_x \in \mathcal{A}_1$ such that $x \in \operatorname{rbd} A_x$. Choose a unit vector v_x pointing from x into A_x. The direction of A_x cannot be parallel to the affine line aff H_x. Otherwise, since x is a relative interior point of H_x, a sufficiently short initial portion of A_x would lie in H_x, contradicting

pairwise disjointness. We call A_x the terminating segment of x. Fix

$$\eta := \frac{1}{10}.$$

For every nonparallel pair consisting of an unoriented host direction and an oriented terminating direction, there exists an invertible linear transformation B which sends the first direction to horizontal and the second direction to upward vertical. By continuity, there exist neighborhoods of those two directions such that, under the same transformation B,

$$|\text{host slope}| \leq \eta,$$

and every terminating direction has positive vertical component and satisfies

$$\left|\frac{dx}{dy}\right| \leq \eta.$$

The space of nonparallel direction pairs is second countable. Hence we may choose countably many such direction charts $(U_k, V_k, B_k), k \geq 1$, covering all possible direction pairs.

Let T_k consist of those $x \in T$ whose host direction and chosen terminating direction belong to the k-th chart. Then $T = \bigcup_{k=1}^{\infty} T_k$. For $x \in T_k$, put

$$p_x := B_k x.$$

Since x is a relative interior point of H_x, there exists $a_x > 0$ such that $B_k H_x$ contains its affine-line portion over every first coordinate within distance a_x of $(p_x)_1$. Similarly, since the transformed terminating direction has positive vertical component, there exists $b_x > 0$ such that $B_k A_x$ extends through every positive vertical displacement less than b_x.

For $m \geq 1$, define

$$T_{k,m} := \left\{ x \in T_k : a_x > \frac{1}{m}, \quad b_x > \frac{1}{m} \right\}.$$

Since $a_x, b_x > 0$,$T = \bigcup_{k,m=1}^{\infty} T_{k,m}$. It therefore suffices to prove

$$\mathcal{L}^{2,*}(T_{k,m}) = 0$$

for every k, m. Fix k, m, put

$$\rho := \frac{1}{m}, \qquad r := \frac{\rho}{10},$$

and work in the transformed coordinates. Partition $\mathbb{R}^2$ into half-open squares of side r. We claim that all points of $B_k(T_{k,m})$ lying in the same square have the same host. Take

$$p = B_k x, q = B_k y, \qquad x, y \in T_{k,m}$$

in the same square. Then $|p_1 - q_1| \leq r, \qquad |p_2 - q_2| \leq r$. Let

$$G_p := B_k H_x, \qquad G_q := B_k H_y.$$

Suppose that $H_x \neq H_y$. Their affine hulls cannot coincide. Indeed, both hosts contain their affine-line portions over horizontal intervals of radius ρ around p and q, respectively, while $|p_1 - q_1| < \rho$. If their affine hulls coincided, the two host portions would overlap, contradicting pairwise disjointness. Thus the two affine hulls are distinct almost-horizontal lines. Over the horizontal interval between p_1 and q_1, both actual host sets are present. They cannot intersect there. Hence one lies strictly above the other. Assume first that the host of q lies above the host of p. Then $G_q(p_1) > p_2$. Since $q \in G_q$ and the slope of G_q has absolute value at most η,

$$0 < G_q(p_1) - p_2 \leq |q_2 - p_2| + \eta |q_1 - p_1| \leq (1 + \eta) r.$$

Follow the terminating member $B_k A_x$ upward from p through vertical distance $3r$. This is possible because $3r < \rho < b_x$. Let the resulting point be z. Then $z_2 = p_2 + 3r$. Since $\left|\frac{dx}{dy}\right| \leq \eta$, we have $|z_1 - p_1| \leq 3\eta r$. Therefore

$$|z_1 - q_1| \leq |z_1 - p_1| + |p_1 - q_1| \leq (1 + 3\eta)r = 1.3r < \rho.$$

Hence the point of the affine line of G_q with first coordinate z_1 actually belongs to G_q.

Moreover,

$$\begin{aligned} G_q(z_1) &\leq G_q(p_1) + \eta|z_1 - p_1| \\ &\leq p_2 + (1 + \eta)r + 3\eta^2 r. \end{aligned}$$

Since $\eta = 1/10$,

$$(1 + \eta) + 3\eta^2 = 1.13 < 3.$$

Thus

$$G_q(z_1) < p_2 + 3r = z_2.$$

At p, the terminating member lies below G_q, while at z it lies above G_q. By continuity, the portion of $B_k A_x$ from p to z intersects G_q. The crossing occurs strictly after p and before vertical displacement $3r < \rho$, so it lies in $B_k A_x$. It also lies within horizontal distance less than ρ of q, so it lies in the relatively open host G_q.

Finally, A_x and H_y must be distinct. If $A_x = H_y$, then p would lie on the affine line of H_y, and since $|p_1 - q_1| < \rho$ while H_y contains its affine-line portion over horizontal radius ρ around q, we would have

$$p \in B_k H_y = B_k A_x,$$

contradicting the fact that x is a relative-boundary point of the relatively open set A_x. Hence two distinct members of the original family intersect, a contradiction. The opposite vertical ordering is treated symmetrically. Therefore all points of $B_k(T_{k,m})$ in one grid square have the same host.

For every grid square Q, there is consequently an affine line L_Q such that

$$B_k(T_{k,m}) \cap Q \subset L_Q.$$

There are countably many grid squares, so $B_k(T_{k,m})$ is contained in a countable union of affine lines. Therefore $\mathcal{L}^{2,*}(B_k(T_{k,m})) = 0$. Since B_k^{-1} is Lipschitz, $\mathcal{L}^{2,*}(T_{k,m}) = 0$. Taking the countable union gives

$$\boxed{\mathcal{L}^{2,*}(T) = 0.}$$

Step 4: A one-sided Klee–Martin lemma.

We prove the following lemma.

Lemma 6 *Let $F \subset \mathbb{R}^2$ be compact. Suppose that for every $x \in F$ there exists a non-degenerate closed segment $S_x = [x, x^*]$ having x as one endpoint, and suppose that $S_x \cap S_y = \varnothing (x \neq y)$ Then $\mathcal{L}^2(F) = 0$.*

Suppose, toward a contradiction, that $\mathcal{L}^2(F) > 0$. For $x \in F$, let $u_x := \frac{x^* - x}{|x^* - x|}$. Fix $\varepsilon := \frac{1}{100}$. For distinct $p, q \in \mathbb{Q}^2$, set $D := |p - q|$ and define

$$W(p, q) := \{x \in F : |x - p| < \varepsilon D, \quad |x^* - q| < \varepsilon D\}.$$

We claim that $F = \bigcup_{\substack{p,q \in \mathbb{Q}^2 \\ p \neq q}} W(p, q)$.

Indeed, fix $x \in F$ and let $\ell := |x^* - x| > 0$. Choose rational p, q sufficiently close to x, x^*, respectively. Then $D = |p - q|$ is arbitrarily close to ℓ, so both inequalities defining $W(p, q)$ hold. Since this is a countable cover and $\mathcal{L}^{2,*}(F) > 0$, there exist p, q such that $a := \mathcal{L}^{2,*}(W(p, q)) > 0$. Fix such a pair and write $W := W(p, q)$.

For every $x \in W$, $|x^* - x| \geq D - |x - p| - |x^* - q| > (1 - 2\varepsilon)D$. Let

$$L := (1 - 2\varepsilon)D.$$

Thus every segment S_x, $x \in W$, has length greater than L. Moreover, $|(x^* - x) - (q - p)| < 2\varepsilon D$. Therefore all the unit directions u_x lie in a sufficiently narrow cone about the direction $q - p$. For $\varepsilon = 1/100$, in particular, $\angle(u_x, u_y) < \frac{\pi}{3} (x, y \in W)$.

Choose $0 < t < \frac{L}{4}$ and define

$$\Phi_t(x) := x + tu_x, \qquad x \in W.$$

We claim that $|\Phi_t(x) - \Phi_t(y)| \geq \frac{1}{4}|x - y| (x, y \in W)$. Take distinct $x, y \in W$, and set

$$u := u_x, \qquad v := u_y, \qquad h := y - x,$$

and

$$w := \Phi_t(y) - \Phi_t(x) = h + t(v - u).$$

Suppose, toward a contradiction, that $|w| < \frac{1}{4}|h|$. If $u = v$, then $w = h$, which is impossible. Hence $u \neq v$. Let $\theta = \angle(u, v)$. Since $\theta < \pi/3$, $|\det(u, v)| = |u - v| \cos\frac{\theta}{2} \geq \frac{1}{2}|u - v|$. Also, $t|u - v| = |w - h| \geq |h| - |w| > \frac{3}{4}|h|$. Hence $|\det(u, v)| > \frac{3}{8t}|h|$.

The two affine lines containing S_x and S_y meet at a unique point, which we write as $x + su = y + rv$. Taking determinants gives

$$s - t = \frac{\det(w, v)}{\det(u, v)}$$

and

$$r - t = \frac{\det(w, u)}{\det(u, v)}.$$

Therefore $|s - t| \leq \frac{|w|}{|\det(u,v)|} < \frac{2}{3}t$, and likewise $|r - t| < \frac{2}{3}t$. Consequently $\frac{t}{3} < s < \frac{5t}{3}$, $\frac{t}{3} < r < \frac{5t}{3}$. Since $t < L/4$, $0 < s, r < L$. But both S_x and S_y have length greater than L, so $x + su = y + rv$ lies in both segments, contradicting pairwise disjointness. Thus $|\Phi_t(x) - \Phi_t(y)| \geq \frac{1}{4}|x - y|$.

Let $W_t := \Phi_t(W)$. Then Φ_t is injective and its inverse $\Phi_t^{-1} : W_t \to W$ is 4-Lipschitz.

We now use the following elementary fact: if g is 4-Lipschitz on a subset of $\mathbb{R}^2$, then

$$\mathcal{L}^{2,*}(g(B)) \leq 128\, \mathcal{L}^{2,*}(B)$$

for every subset B of its domain. Indeed, cover B by squares of side lengths s_i. The image of the part lying in one such square has diameter at most $4\sqrt{2}\, s_i$, and can therefore be covered by a square of area at most $128 s_i^2$. Taking infima over all square covers gives the estimate.

Applying this to Φ_t^{-1} yields $a = \mathcal{L}^{2,*}(W) \leq 128\, \mathcal{L}^{2,*}(W_t)$. Hence $\mathcal{L}^{2,*}(W_t) \geq \frac{a}{128}$. Now define

$$N_t(F) := \{z \in \mathbb{R}^2 : \operatorname{dist}(z, F) \leq t\}.$$

Since $|\Phi_t(x) - x| = t$, we have $W_t \subset N_t(F)$. Moreover, $W_t \cap F = \varnothing$. Indeed, if $z = \Phi_t(x) \in F$, then $z \neq x$ and $z \in S_x$. Since $z \in F$, there is also a segment S_z rooted at z, and hence $z \in S_x \cap S_z$, contradicting pairwise disjointness.

Therefore $W_t \subset N_t(F) \setminus F$. Since F is compact, $N_{1/n}(F) \downarrow F (n \to \infty)$. The sets $N_{1/n}(F)$ have finite measure, so continuity of Lebesgue measure from above gives $\mathcal{L}^2\big(N_{1/n}(F) \setminus F\big) \longrightarrow 0$. Choose n sufficiently large that $t := \frac{1}{n} < \frac{L}{4}$ and $\mathcal{L}^2\big(N_t(F) \setminus F\big) < \frac{a}{128}$. Then $\frac{a}{128} \leq \mathcal{L}^{2,*}(W_t) \leq \mathcal{L}^2\big(N_t(F) \setminus F\big) < \frac{a}{128}$, a contradiction. Therefore $\mathcal{L}^2(F) = 0$.

Step 5: Completion of the theorem.

Suppose, toward a contradiction, that $\mathcal{L}^2(\mathcal{V}) > 0$. We have shown that $\mathcal{L}^2(P) = 0$ and $\mathcal{L}^2(T) = 0$. Set $R := \mathcal{V} \setminus (P \cup T)$. Since $\mathcal{V}$ is Lebesgue measurable, P is Borel, and T is null and hence Lebesgue measurable, the set R is Lebesgue measurable. Moreover,

$$\mathcal{L}^2(R) = \mathcal{L}^2(\mathcal{V}) > 0.$$

Since $K = P \cup E$, every point of R belongs to E, and hence $R \subset E \setminus T$. By inner regularity of Lebesgue measure, there exists a compact set $F \subset R$ such that $\mathcal{L}^2(F) > 0$.

For every $x \in F$, choose one A_x satisfying $x \in \operatorname{rbd} A_x$. Choose a nondegenerate closed segment S_x starting at x and extending inward along A_x, so that $S_x \setminus \{x\} \subset A_x$. If A_x is a bounded interval of length ℓ_x, choose $\operatorname{length}(S_x) < \frac{\ell_x}{3}$. If A_x is a ray, choose any finite positive length.

We claim that $S_x \cap S_y = \varnothing (x \neq y)$. If $A_x = A_y$, then that common member must be a bounded interval whose two endpoints are x and y. Since both rooted segments have length less than one third of the full interval length, they are disjoint. Now suppose $A_x \neq A_y$. If $z \in S_x \cap S_y$ and $z \neq x, y$, then $z \in A_x \cap A_y$, contradicting pairwise disjointness of the original family. If $z = x$, then $x \in S_y \setminus \{y\} \subset A_y$. Since x is an endpoint of A_x, it follows that x is a T-junction: $x \in T$, contrary to $F \subset R$. The case $z = y$ is identical. Therefore $S_x \cap S_y = \varnothing (x \neq y)$. Thus F is a compact set of distinguished endpoints of a pairwise disjoint family of nondegenerate closed line segments. By the one-sided Klee–Martin lemma, $\mathcal{L}^2(F) = 0$, contradicting $\mathcal{L}^2(F) > 0$. Hence $\mathcal{L}^2(K) = 0$. ∎

Example 3 *We want to find an example of a convex partition preserved by a particular martingale coupling, [9] and this partition is strictly finer than the irreducible convex paving.[10] If such an example exists, then the finest convex partition preserving the convex-order relationship within each cell can be strictly finer than the irreducible convex paving introduced by De March and Touzi (2019). The following example demonstrates that a convex partition preserved by a particular martingale coupling can indeed be strictly finer than the irreducible convex paving.*

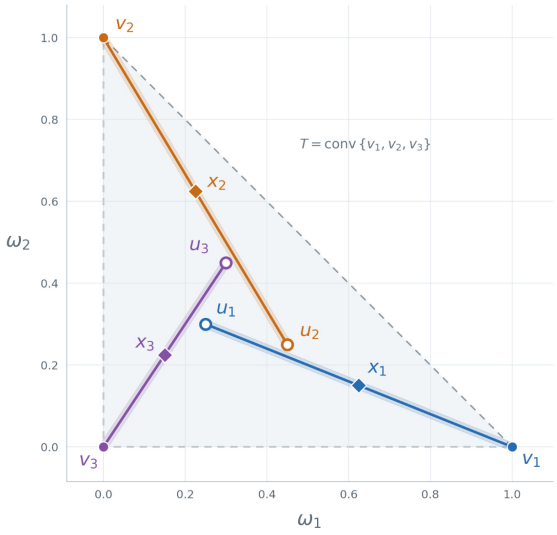
Consider the points

$$v_1 = (1, 0), \qquad v_2 = (0, 1), \qquad v_3 = (0, 0),$$
$$u_1 = \left(\frac{1}{4}, \frac{3}{10}\right), \quad u_2 = \left(\frac{9}{20}, \frac{1}{4}\right), \quad u_3 = \left(\frac{3}{10}, \frac{9}{20}\right).$$

[9] It means that there exists a feasible martingale transport that maps each point only to points within the same convex cell.

[10] The irreducible paving is the $\{A_x\}_{x \in X}$ defined in Proposition 1.

Figure 3: Example 3



For $i = 1, 2, 3$, set

$$x_i = \frac{v_i + u_i}{2}, \qquad S_i = [v_i, u_i], \qquad T = \mathrm{conv}\{v_1, v_2, v_3\}.$$

Let $r = 1/100$, let $B_r(z)$ denote the open Euclidean ball of radius r centered at z, and define the open convex tubes

$$C_i = S_i + B_r(0).$$

A direct calculation gives $\min_{i \neq j} \mathrm{dist}(S_i, S_j) = \frac{3}{20\sqrt{34}} > 2r$.

Hence the tubes are pairwise disjoint, and $\mathcal{C} = \{C_1, C_2, C_3\} \cup \big\{\{z\} : z \in \mathbb{R}^2 \setminus \bigcup_{i=1}^{3} C_i\big\}$ *is a Borel measurable partition into relatively open convex sets. Write $\mathcal{C}(x)$ for the cell containing x. For $z \in \mathbb{R}^2$, let η_z be the uniform probability measure on $B_r(z)$. Its barycenter is z. Define*

$$\mu = \frac{1}{3} \sum_{i=1}^{3} \delta_{x_i}, \qquad \nu = \frac{1}{6} \sum_{i=1}^{3} \big(\eta_{v_i} + \eta_{u_i}\big).$$

In particular, ν is absolutely continuous with respect to two-dimensional Lebesgue measure.

Define a coupling P by

$$P(dx, dy) = \frac{1}{3}\sum_{i=1}^{3} \delta_{x_i}(dx) P_{x_i}(dy), \qquad P_{x_i} = \frac{1}{2}\eta_{v_i} + \frac{1}{2}\eta_{u_i}.$$

Its marginals are μ and ν, and

$$\int_{\mathbb{R}^2} y \, P_{x_i}(dy) = \frac{v_i + u_i}{2} = x_i.$$

Therefore $P \in \mathcal{M}(\mu, \nu)$, and in particular $\mu \preceq_{\mathrm{cx}} \nu$. Since both $B_r(v_i)$ and $B_r(u_i)$ lie in C_i,

$$P_{x_i}(C_i) = 1 \quad \textit{for every } i, \qquad P\big(Y \in C(X)\big) = 1.$$

Thus this coupling transports each source entirely within its own cell.

To identify the irreducible paving, construct another coupling Q. Set

$$Q_{x_i} = \frac{1}{2}\eta_{u_{i+1}} + \frac{2}{5}\eta_{v_i} + \frac{1}{40}\eta_{v_{i+1}} + \frac{3}{40}\eta_{v_{i+2}} \quad \textit{for } i = 1, 2, 3,$$

and

$$Q(dx, dy) = \frac{1}{3}\sum_{i=1}^{3} \delta_{x_i}(dx) Q_{x_i}(dy).$$

The coordinates above satisfy $\frac{1}{2}u_{i+1} + \frac{2}{5}v_i + \frac{1}{40}v_{i+1} + \frac{3}{40}v_{i+2} = x_i, \qquad i = 1, 2, 3.$ Moreover, $\frac{1}{3}\sum_{i=1}^{3} Q_{x_i} = \frac{1}{6}\sum_{i=1}^{3}\big(\eta_{u_i} + \eta_{v_i}\big) = \nu$. Consequently $Q \in \mathcal{M}(\mu, \nu)$.

Every Q_{x_i} assigns positive mass to the disks around all three vertices. Since $u_1, u_2, u_3 \in T$, it follows that

$$\overline{\mathrm{conv}}\big(\mathrm{supp}\, Q_{x_i}\big) = T + \overline{B_r(0)} = \overline{\mathrm{conv}}\big(\mathrm{supp}\, \nu\big), \qquad i = 1, 2, 3.$$

For any $R \in \mathcal{M}(\mu, \nu)$, each R_{x_i} is absolutely continuous with respect to ν, because $\mu(\{x_i\}) > 0$. Thus no such conditional distribution has a larger closed convex support than Q_{x_i}. By

the maximal-support characterization of the irreducible paving in Theorem 2.1 De March and Touzi (2019),

$$I(x_1) = I(x_2) = I(x_3) = \mathrm{ri}\big(T + \overline{B_r(0)}\big) = T + B_r(0).$$

In particular, $C_i \subsetneq I(x_i), \qquad i = 1, 2, 3$. *Every remaining cell of* C *is a singleton and therefore lies in a cell of the irreducible paving. Hence* C *strictly refines that paving.*

Proof of Proposition 3. Put $I = \mathrm{ri}\, X$, $L = \mathrm{span}(X - X)$, and $r = \dim L$. By the concentration hypothesis, $\mu(I) = 1$ and the canonical conditional closed convex support is X, μ-almost everywhere. In particular, $X = \mathrm{conv}(\mathrm{supp}\, \mu_0)$: one inclusion follows from the second marginal constraint and the canonical support property, and the other from $\mu_0(X) = 1$ and compactness of X. The case $r = 0$ is immediate.

Step 1: An affine-annihilating perturbation. Suppose that $\mathrm{supp}\, \mu$ is not affinely independent. We construct a nonzero bounded Borel function h, supported on a compact set $K \subset I$, such that

$$\int_X h \, d\mu = 0, \qquad \int_X x h(x) \, d\mu(x) = 0. \tag{3}$$

If $\mathrm{supp}\, \mu = \{x_1, \ldots, x_m\}$ is finite, all its points belong to I. Choose coefficients b_i, not all zero, with $\sum_i b_i = 0$ and $\sum_i b_i x_i = 0$, and set $h(x_i) = b_i / \mu(\{x_i\})$, with $h = 0$ elsewhere and $K = \{x_1, \ldots, x_m\}$.

If the support is infinite, choose $r + 2$ distinct points in $I \cap \mathrm{supp}\, \mu$ and pairwise disjoint relatively open neighborhoods B_i whose closures are compact subsets of I. Such points exist because $\mu(I) = 1$. Each $\mu(B_i) > 0$. The $r + 2$ vectors

$$v_i = \left(\mu(B_i), \int_{B_i} x \, d\mu(x) \right)$$

lie in the $(r+1)$-dimensional space $\mathrm{span}\{(1, x) : x \in X\}$ and are therefore linearly dependent. Choose coefficients a_i, not all zero, with $\sum_i a_i v_i = 0$. Taking $h = \sum_i a_i \mathbf{1}_{B_i}$ and $K = \bigcup_i \overline{B_i}$

gives (3). In either case, put

$$\sigma = h\mu, \qquad H = \int_X |h| \, d\mu > 0.$$

Then σ is nonzero, is supported on $K \subset I$, and annihilates every affine function.

Step 2: Robust anchor neighborhoods. There are finitely many points $y_1, \ldots, y_N$ in $S = \operatorname{supp} \mu_0$ such that

$$P_0 = \operatorname{conv}\{y_1, \ldots, y_N\}, \qquad \operatorname{aff} P_0 = \operatorname{aff} X, \qquad K \subset \operatorname{ri} P_0.$$

Indeed, for each $x \in K$, take an r-dimensional simplex $\Delta_x \subset I$ containing x in its relative interior. Its vertices are finite convex combinations of points of S, since $X = \operatorname{conv} S$. A finite subcover of K by the sets $\operatorname{ri} \Delta_x$ yields the required finite set of anchors. Choose $\delta > 0$ with $K + \delta B_L \subset P_0$, where $B_L = \{v \in L : \|v\| \leq 1\}$, and set

$$U_j = \{y \in X : \|y - y_j\| < \delta/2\}.$$

Every U_j has positive μ_0-measure. If $z_j \in U_j$ for all j, the support functions satisfy, for every unit vector $u \in L$,

$$\max_j u \cdot z_j \geq \max_j u \cdot y_j - \delta/2 \geq \sup_{z \in K} u \cdot z + \delta/2.$$

Consequently,

$$K \subseteq \operatorname{conv}\{z_1, \ldots, z_N\} \quad \text{whenever } z_j \in U_j \text{ for all } j. \tag{4}$$

Step 3: Uniformly positive anchor probabilities. By Corollary 2, there is $P \in \mathcal{M}(\mu, \mu_0)$ whose conditional distributions have support S, μ-almost everywhere. In particular, $P_x(U_j) > 0$ for all j, simultaneously for almost every x. For a Borel version of this

kernel define

$$E_m = \{x \in I : P_x(U_j) \geq 1/m \text{ for all } j\}.$$

Since $\mu(\bigcup_m E_m) = 1$, some $E = E_{m_0}$ has positive μ-measure. Write $c = 1/m_0$, so that

$$P_x(U_j) \geq c \quad \text{for all } j \text{ and } \mu\text{-almost every } x \in E. \tag{5}$$

Step 4: A uniform bound by the convex-order slack. Fix a continuous convex function $\varphi : X \to \mathbb{R}$ and let

$$g_\varphi(x) = \int_X \varphi(y) P_x(dy) - \varphi(x).$$

Jensen's inequality and the marginal constraints give $g_\varphi \geq 0$ almost everywhere and

$$\int_X g_\varphi \, d\mu = \int_X \varphi \, d\mu_0 - \int_X \varphi \, d\mu. \tag{6}$$

For $x \in E \subset I$ outside the exceptional null set, choose a relative subgradient $\xi_x \in L$ and define

$$R_x(z) = \varphi(z) - \varphi(x) - \xi_x \cdot (z - x) \geq 0, \qquad z \in X.$$

The subgradient exists because $x \in \operatorname{ri} X$; its supporting inequality extends to the boundary by continuity. The martingale property gives $\int_X R_x \, dP_x = g_\varphi(x)$. Thus, for each $\eta > 0$ and each j, there is $z_j \in U_j$ such that

$$R_x(z_j) \leq \frac{g_\varphi(x)}{c} + \eta,$$

because the average of R_x under the normalized restriction of P_x to U_j is at most $g_\varphi(x)/c$. By (4) and convexity of R_x,

$$\sup_{z \in K} R_x(z) \leq \max_j R_x(z_j) \leq \frac{g_\varphi(x)}{c} + \eta.$$

Letting $\eta \downarrow 0$ and using the affine-annihilation property of σ yields

$$\left|\int_X \varphi \, d\sigma\right| = \left|\int_X R_x \, d\sigma\right| \le H \sup_{z\in K} R_x(z) \le \frac{H}{c}\, g_\varphi(x).$$

Integrating over E and using (6), we obtain

$$\int_X \varphi \, d\mu_0 - \int_X \varphi \, d\mu \ge \kappa \left|\int_X \varphi \, d\sigma\right|, \qquad \kappa = \frac{c\mu(E)}{H} > 0. \tag{7}$$

The constant is independent of φ. No measurable choice of ξ_x is needed: only the measurable function g_φ is integrated.

Step 5: Two distinct feasible perturbations. Choose $0 < \varepsilon < \min\{\kappa, \|h\|^{-1}_{L^\infty(\mu)}\}$ and define $\mu^\pm = (1 \pm \varepsilon h)\mu$. They are probability measures with the same barycenter as μ. For every continuous convex φ,

$$\int_X \varphi \, d\mu_0 - \int_X \varphi \, d\mu^\pm \ge (\kappa - \varepsilon) \left|\int_X \varphi \, d\sigma\right| \ge 0.$$

In particular this inequality holds for restrictions to X of all finite convex functions on $\mathbb{R}^n$, so $\mu^\pm \in MPC(\mu_0)$. Since $\sigma \neq 0$, $\mu^+ \neq \mu^-$ and $\mu = (\mu^+ + \mu^-)/2$. Thus μ is not extreme.

■

Given two probability measure $\mu_1, \mu_2 \in MPC(\mu_0)$, we say μ_1 has finer concentration regions than μ_2 with respect to μ_0 if for the corresponding $\{A^1_x\}_{x\in X}$ and $\{A^2_x\}_{x\in X}$ defined in Proposition 1, we have $A^1_x \subset A^2_x$ for μ_1-a.s. all x.

Lemma 7 *Given two probability measures $\mu_1, \mu_2 \in MPC(\mu_0)$, if $\mu = \alpha\mu_1 + (1-\alpha)\mu_2, \alpha \in (0,1)$, then both μ_1 and μ_2 have finer concentration regions than μ with respect to μ_0.*

Proof of Lemma 7. Let $\hat{p}^i \in \mathcal{M}(\mu_i, \mu_0)$, $i = 1, 2$, be the couplings that induce $\{A^i_x\}_{x\in X}$ as in Proposition 1. Since $\mu = \alpha\mu_1 + (1-\alpha)\mu_2$, the mixture $q := \alpha\hat{p}^1 + (1-\alpha)\hat{p}^2$ belongs to

$\mathcal{M}(\mu, \mu_0)$. Its conditional kernel is

$$q_x = \alpha \frac{d\mu_1}{d\mu}(x)\hat{p}^1_x + (1-\alpha)\frac{d\mu_2}{d\mu}(x)\hat{p}^2_x \qquad \text{for } \mu\text{-almost every } x.$$

The coefficient of $\hat{p}^i_x$ is positive for μ_i-almost every x. Hence, by Proposition 1, the maximal coupling $\hat{p} \in \mathcal{M}(\mu, \mu_0)$ satisfies

$$C^i_x := su\hat{p}p(\hat{p}^i_x) \subseteq su\hat{p}p(q_x) \subseteq C_x := su\hat{p}p(\hat{p}_x) \qquad \text{for } \mu_i\text{-almost every } x,\ i = 1, 2.$$

By Proposition 1, $x \in \operatorname{ri} C^i_x \cap \operatorname{ri} C_x$. For any $y \in \operatorname{ri} C^i_x$ with $y \neq x$, the ray from x through y extends beyond y to some $z \in C^i_x \subseteq C_x$. Since $x \in \operatorname{ri} C_x$, the segment $[x, z)$ lies in $\operatorname{ri} C_x$. Thus $y \in \operatorname{ri} C_x$, and $A^i_x = \operatorname{ri} C^i_x \subseteq \operatorname{ri} C_x = A_x$ for μ_i-almost every x, $i = 1, 2$.

■